JCUIScaler – RoadMap

Rev I – status as of 2026-11-08 (ready for the 1.0 publication; document closed for 1.0)

Project status

Current status	**RC2 completed; RC3 in progress — project ready for the 1.0 publication**
Completed	Technical Specification R1-09 Static Analysis Rev L Risk Analysis Rev E (RC1.2, RC2 and 1.0 updates) RC1 code cleanup (Static Analysis Rev D → Rev F) Demo project (manual integration, PageControl, grids, ListView, TreeView) RC1.2 — source-level Delphi compatibility (unofficial) + bug fixes (Rev G) Demo — rebuild and GTK2 runtime tests (R1-05, Rev H) RC2 — TJCScaler component, Lazarus packages, license (R2-02, R2-04, R2-05) English documentation and README PL/EN (R2-03); repository structure (R3-01) Delphi 13.2 (Win64) verification: compilation and basic operation macOS 27 verification (Lazarus 4.8, Cocoa) CONTRIBUTING, issue templates, CI (GitHub Actions)
Next stage	RC3 — public repository (R3-02), OPM package and submission (R3-04, R3-05)

After the cleanup session the RC1 code is treated as a stable reference implementation. All static analysis findings have been resolved. The library is functionally complete for version 1.0 with one exception: monitor-change handling (R2-01) is the last required functional item before RC3. RC1.2 (2026-11-06) added unofficial source-level compatibility with Delphi VCL and fixed three bugs found while preparing it (BorderSpacing, GTK2 compilation, rounding of negative values).

Decision 2026-11-07: monitor-change handling (R2-01) is moved beyond version 1.0 and is a documented limitation in 1.0. The IDE component (R2-02) remains in the 1.0 scope.

RC2 (2026-11-07): the version 1.0 code is closed — the TJCScaler component (R2-02), the LCL-compatible license (R2-04) and the attribution files (R2-05) are complete. The TJCUIScaler core is unchanged in code since RC1.2. English documentation (R2-03) completed the same day; the publication stage (RC3) remains.

After the 1.0 publication, further planning is tracked in the project repository (GitHub: issues, milestones). Tasks open to the community ("help wanted"): R2-01 — monitor change with a different DPI (LCL), DPI-change handling in Delphi/VCL, and font scaling (experimental). This RoadMap remains the record of the plan up to version 1.0 and is not updated on an ongoing basis; the current document is the Technical Specification.

RC1 / RC1.1 / RC1.2 – Core stabilisation

Goal: complete the core functionality. All RC1.1 items were completed in the cleanup session of 2026-06-01. RC1.2 (R1-04) completed 2026-11-06.

ID	Scope	Status
R1-01	**ResetOnShow — restoring the initial state on OnShow** – Implementation of RestoreInitialScale (brute-force proportional restore). – New property ResetOnShow: Boolean (default False). – Documented limitation: does not cover Minimize/Restore and Maximize/Restore.	COMPLETED
R1-02	**RC1 code cleanup based on Static Analysis Rev D** – LocalRound as a private method of the object — uniform rounding throughout the code. – Find* for ListView, StatusBar, TreeView — unified lookups. – JCSCALER_BORDER_TAG as a public constant in the interface section. – SavedBevel/SavedBorder initialised at declaration. – GridLastWOffset — kept, with an extended comment. – TGroupBox — registration kept, section in ApplyControlExceptions deferred. – Code comments unified in English.	COMPLETED
R1-03	**Demo project** – Manual integration — three calls in three events. – PageControl with three tabs. – StringGrid, ListView, TreeView (3 nesting levels). – GroupBox with form controls. – StatusBar with panels. – Second form (TJCUITestForm) — multi-form test. – ResetOnShow = True on both forms.	COMPLETED
R1-04	**RC1.2 — source-level Delphi compatibility + bug fixes** – Source-level compatibility with Delphi VCL (unofficial) — manual integration only, no Delphi package. – Object architecture unchanged; three {$IFDEF FPC} blocks (header, BorderSpacing capture and apply). – BorderSpacing bug fix: capture stored the control position instead of BorderSpacing; apply was unreachable for grids, ListView, TreeView, StatusBar and Splitter. – LCLGTK2 bug fix: DefaultItemHeight instead of the non-existent ItemHeight — RC1 did not compile for GTK2. – LocalRound: Floor(V + 0.5) — correct rounding of negative values. – File names: JCUIScaler.capture.inc, JCUIScaler.controls.inc, JCUIScaler.find.inc. – Specification correction: TSpeedButton is not excluded; the	COMPLETED

	reference model is built on the first ScalerOnShow. – Verification: FPC 3.2.2 / LCL 3.0 without warnings; Delphi 13.2 compilation confirmed (after the TSplitter fix, 2026-11-07).	
R1-05	**Demo — rebuild and GTK2 runtime tests** – source/ + demo/ structure; the demo takes the library from ../source (no code copies). – Default anchors instead of Anchors = [] — correct layout at a DPI different from DesignTimePPI. – StatusBar with panels, panel with JCSCALER_BORDER_TAG on the second form, English texts. – ScrollBars = ssVertical for ListView and TreeView — recommended configuration (Spec section 13.1). – Demo for Lazarus only; only the three standard calls, no component. – GTK2 tests (Xvfb) at 96 and 120 DPI: no drift, all tabs, second form. – Library code unchanged (identical to RC1.2).	**COMPLETED**

RC2 – Integration layer, license and English documentation

Goal: simplify integration with an IDE component (R2-02, completed), license and attribution (R2-04, R2-05, completed) and English documentation (R2-03). Monitor-change handling (R2-01) moved beyond 1.0.

ID	Scope	Status
R2-01	**Monitor-change handling — MOVED BEYOND 1.0** – Scenario: Form.Scaled = True + moving the window between monitors. – LCL rescales the form automatically → OnResize fires on an already correct state. – Required reaction: accept the new state as the reference, do not revert it. – Implementation hypothesis: MonitorChanged → CaptureOriginals → skip one ScaleForm. – Mechanism: flag FSkipNextResize set before CaptureOriginals, cleared in ScaleForm. – Challenge: monitor-change detection differs between LCL widgetsets. – Challenge: distinguishing a resize caused by a monitor change from a manual resize by the user. – Challenge: the order of LCL events (end of scaling vs OnResize) may be platform-dependent. – Required tests: Windows (multi-monitor), Linux/GTK2, macOS. – Risk: a deterministic implementation may not be possible on all platforms. – Status after 1.0: a “help wanted” issue in the project repository (together with DPI-change handling in Delphi/VCL).	**AFTER 1.0**

	– Delphi: outside the 1.0 scope. VCL provides OnBeforeMonitorDpiChanged / OnAfterMonitorDpiChanged — possible future support following the same hypothesis. – Decision 2026-11-07: beyond 1.0. Documented limitation in 1.0: Form.Scaled + moving the window between monitors with different DPI is not supported.	
R2-02	**TJCScaler — IDE component** – Integration layer — does not replace the TJCUIScaler core; calls only its public methods. – Class TJCScaler (instance JCScaler1); the core keeps the name TJCUIScaler (compatibility with production applications). – Automatic integration: ScalerInit in Loaded, OnShow chained, OnResize via AddHandlerOnResize. – Properties SkipBevelScaling and ResetOnShow linked to the core object. – Events BeforeShow, AfterShow, BeforeResize, AfterResize; virtual methods DoShow and DoResize. – Packages jcuiscaler_rt (RunTime) and jcuiscaler_dsgn (DesignTime); palette icons 24/36/48 px; palette page JCUI. – Tests: 12/12 screenshots identical to manual integration (GTK2); IDE installation confirmed on Windows (Lazarus 4.8). – LCL only. The demo intentionally does not use the component.	**COMPLETED**
R2-03	**English documentation** – Translations: Technical Specification, Static Analysis, Risk Analysis, RoadMap — docs/en (DOCX and PDF). – README.md (EN) and README.pl.md (PL): description, quick start (component and three calls), FAQ, limitations, license. – The Polish version remains the source document and prevails in case of discrepancies.	**COMPLETED**
R2-04	**License** – License identical to the LCL: GNU Library General Public License, version 2, with the static linking exception (modified LGPL). – COPYING.LGPL.txt and the exception text in COPYING.modifiedLGPL.txt — identical to the Lazarus files. – Correction: the earlier entry “LGPL 2.1” was wrong — the LCL license is LGPL version 2.	**COMPLETED**
R2-05	**Licensing and attribution** – LICENSE, COPYRIGHT, AUTHORS files in the root directory. – License headers in all source files (Copyright (c) 2025-2026 Jakub Cebula, Kraków). – Preserving author information: the copyright notice and the original author's name must not be removed from the source files — stated in the headers and in LICENSE; the license itself is unchanged. The notice travels with the files also with manual integration and distribution via OPM. – License field in the .lpk packages updated.	**COMPLETED**

NOTE: R2-01 moved beyond 1.0 (decision 2026-11-07); in 1.0 monitor-change handling is a documented limitation. The IDE component (R2-02) was completed on 2026-11-07 — the version 1.0 code is closed; English documentation (R2-03) completed — RC2 closed entirely.

RC3 – Preparing the publication

Goal: prepare the project for public release. RC3 starts after RC2 is completed.

ID	Scope	Status
R3-01	**Repository structure** – /source — library and component code; /source/design — IDE registration and icons. – /packages/lazarus — packages jcuiscaler_rt.lpk and jcuiscaler_dsgn.lpk. – /demo — demo project (Lazarus, manual integration). – /docs/pl, /docs/en — documentation. – /assets — icons and logo. – Root directory: README (PL, EN), LICENSE, COPYING.LGPL.txt, COPYING.modifiedLGPL.txt, COPYRIGHT, AUTHORS, HISTORY.txt.	COMPLETED
R3-02	**Public repository** – Decision (2026-11-08): GitHub — the main repository, where issues and pull requests are accepted.	PLANNED
R3-03	**README** – Project description and motivation. – Quick start (3 calls, ready-to-use code). – Supported components and limitations. – Delphi (unofficial) — scope, manual integration, limitations. – FAQ. – FAQ: anchors, TListView scrollbars (ssVertical), hidden tab sheets, GTK2 ssBoth.	COMPLETED
R3-04	**OPM package** – Package file (.lpk). – Package metadata and description. – Preparing the submission. – .lpk packages ready (version 1.0.0); to do: OPM “Create repository package” and submission metadata.	PLANNED
R3-05	**Submission to OPM** – Publishing the package for the Lazarus community.	PLANNED

Conditions for version 1.0

Area	Conditions
Code	✔ RC1 completed and stabilised (RC1.2) ✔ RC2 — code complete (R2-02, R2-04, R2-05; R2-01 beyond 1.0) ○ RC3 completed
Documentation	✔ PL complete (Spec R1-09, Static Analysis Rev L, RoadMap Rev I, Risk Analysis Rev E); documents also as PDF; supplementary documents in docs/*/Additional ✔ EN complete (R2-03); README PL/EN
Distribution	○ Public repository (R3-02) ○ OPM package submitted (R3-05)

Result: JCUIScaler 1.0 — a publicly available project distributed via the Lazarus OPM.

RoadMap revision history

- Rev A — initial version.
- Rev B — update after completing RC1.
- Rev C (2026-06-01) — update after the RC1 cleanup: R1-01/R1-02/R1-03 closed, R2-01 extended with the monitor-change implementation hypothesis, R2-01 marked as the last functional item before 1.0, status of the Polish documentation updated.
- Rev D (2026-11-06) — R1-04 added (RC1.2: source-level Delphi compatibility, BorderSpacing/GTK2/LocalRound bug fixes, .inc files renamed), Delphi note in R2-01, references updated to Spec R1-04 and Static Analysis Rev G.
- Rev E (2026-11-07) — R2-01 moved beyond 1.0, R2-02 confirmed for 1.0 (hook mechanism described), R1-05 added (demo rebuild, GTK2 runtime tests), references updated to Spec R1-05 and Static Analysis Rev H.
- Rev F (2026-11-07) — RC2: R2-02 (TJCScaler component), R2-04 (license — corrected to LGPL v2 as in the LCL) and R2-05 (attribution files, headers) completed; repository structure updated (R3-01); references to Spec R1-06, Static Analysis Rev I and Risk Analysis Rev C.
- Rev G (2026-11-07) — R2-03 (English documentation, README) completed; R3-01 (repository structure) and R3-03 (README) completed; references updated to Spec R1-07 and Static Analysis Rev J; next stage: public repository and OPM.
- Rev H (2026-11-07) — Delphi 13.2 verification (compilation, basic operation, third-party controls: TMS UI Pack PageControl, RxDBGrid); TSplitter fix (F-28) and CRLF line endings (F-29); references updated to Spec R1-08, Static Analysis Rev K and Risk Analysis Rev D; only the RC3 stage remains before publication.
- Rev I (2026-11-08) — document closed for version 1.0: macOS 27 verification (Lazarus 4.8, Cocoa); CONTRIBUTING, issue templates and CI; “help wanted” tasks (R2-01, DPI in Delphi/VCL, font scaling — experimental); further planning in the project repository; document moved to docs/*/Additional.